GE VERNOVA

PRODUCT SERVICE

TIL 1945-R2
Publish Date: 15 JULY 2024
Compliance Category - **A**
Timing Code - **6**

TECHNICAL INFORMATION LETTER

F-CLASS TURBINE WHEEL INSPECTION AND MAINTENANCE RECOMMENDATIONS

APPLICATION
Select F-class gas turbine rotors.

PURPOSE
Advise users to perform routine inspections at a maintenance interval based on installed configuration. This TIL supersedes TILs 1280, 1327, 1434, 1539, 1540, and 1742.

REASON FOR REVISION
1) Added a new 3 Step-Age Heat Treatment category for Stage 1 & Stage 2 wheels applicable to 7F and 6F.

2) Added a specific ECI inspection interval for Stage 1 contoured and enhanced wheels with 2 Step-Age heat treatment,

3) Added a specific ECI inspection interval for Stage 2 wheels post Rotor Life Extension.

COMPLIANCE CATEGORY

M - Maintenance	Identifies maintenance guidelines or best practices for reliable equipment operation.
C - Compliance Required	Identifies the need for action to correct a condition that, if left uncorrected, may result in reduced equipment reliability or efficiency. Compliance may be required within a specific operating time.
A - Alert	Failure to comply with the TIL could result in equipment damage or facility damage. Compliance is mandated within a specific operating time.
S - Safety	Failure to comply with this TIL could result in personal injury. Compliance is mandated within a specific operating time.

TIMING CODE

1	Prior to Unit Startup / Prior to Continued Operation (forced outage condition)
2	At First Opportunity (next shutdown)
3	Prior to Operation of Affected System
4	At First Exposure of Component
5	At Scheduled Component Part Repair or Replacement.
6	Next Scheduled Outage

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

BACKGROUND DISCUSSION

The function of the turbine rotor in a heavy-duty gas turbine engine includes carrying turbine blade loads, transmitting torque to the compressor and generator for power generation, as well as supplying the turbine blades with cooling air. Prior to assembly, the individual disks or wheels are balanced by the addition of weights installed in a groove. The weights are held in place by deforming the metal on either the balance weight or the balance weight groove. The turbine blades are retained axially by a snap ring type lock wire that is located in a machined groove passing through the turbine blades and turbine wheel. The portion of the wheel retaining the lock wire is known as the lock wire tab. The F-class turbine blade cooling air supply is provided through cooling slots in the bottom of the stage 1 and 2 turbine wheel dovetails.

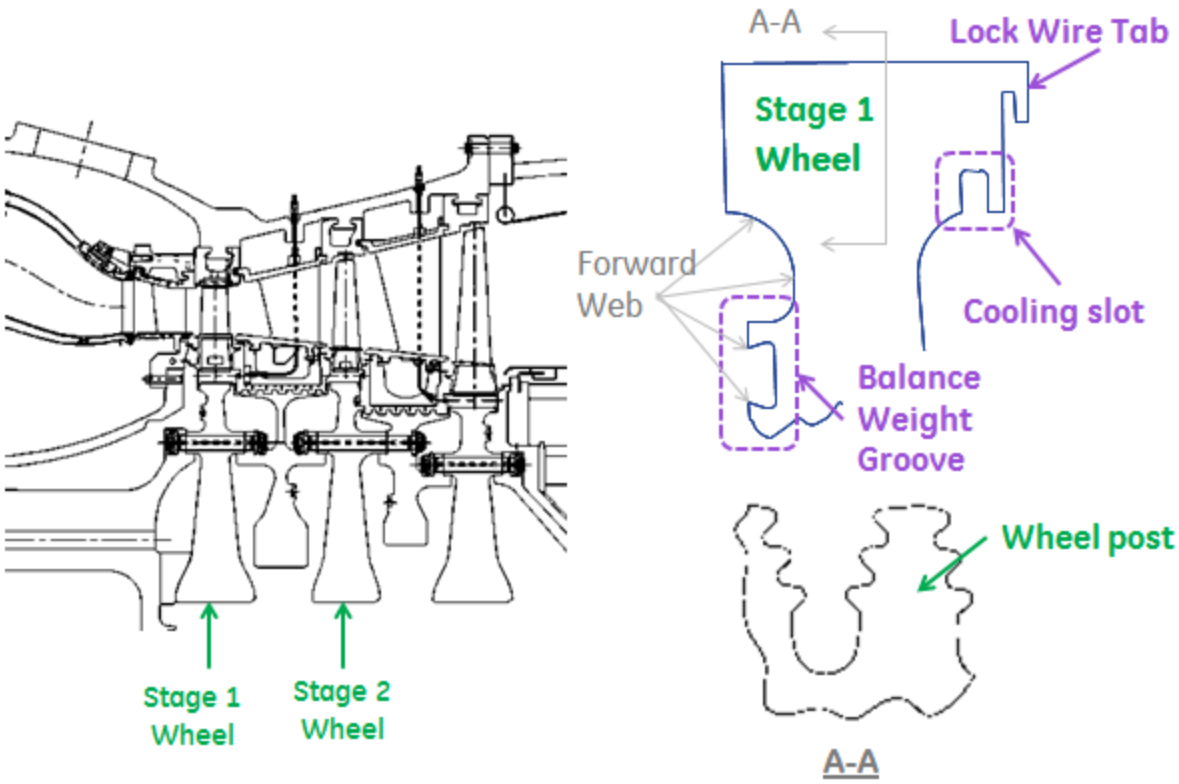


Figure 1: F-class Turbine Rotor Features

The disks in the turbine rotor are made of a high nickel alloy that is sensitive to surface imperfections where an intergranular crack may initiate and propagate due to high stresses and temperatures. Depending on the initiation location, an undetected crack may continue to propagate due to exposure to stress and temperature until either:

1- the crack is detected and either repaired or the rotor is removed from service, or

2- the crack propagates to a critical length, releasing a wheel section and/or adjacent turbine blades into the turbine hot gas path section potentially resulting in significant damage.

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Continuous improvements have been made to the turbine rotor wheels over time in order to improve their overall damage sensitivity.

Rotors operating in high ambient and humid or corrosive conditions have an increased risk of crack initiation and propagation. Recommended inspections and inspection intervals to help mitigate crack propagation are provided based on wheel features, such as: shot peening condition at first fire, cooling slot geometry, material heat treatment, and environmental conditions. In some cases, rotors may be swapped from site to site in order to expedite an outage. This TIL is being issued to all general F-Class turbine rotors in order to provide recommendations for the various configurations. Rotors operating in high ambient and/or corrosive conditions have an increased risk of crack initiation and propagation and are addressed in TIL 1937.

Shot Peening Effect

Early generation turbine rotors have the highest sensitivity to surface imperfections. These rotors were manufactured and fired without full shot peen coverage of the turbine wheels. Shot peening leaves a residual compressive stress and decreases the risk of initiating a crack. Wheels that operated without shot peening have an increased risk of initiating and propagating a crack in the rim region of the turbine rotor and in the turbine wheel dovetail slots. After 1996, all F-class turbine wheels shipped with full peening coverage; these wheels were never fired without shot peening.

Cooling Slot

During operation, the intersection of the cooling slot and dovetail slot is subjected to high temperatures and tensile stresses caused by turbine blade pull. The high temperatures and stresses in conjunction with an adverse edge condition have been known to initiate cracks in this location.

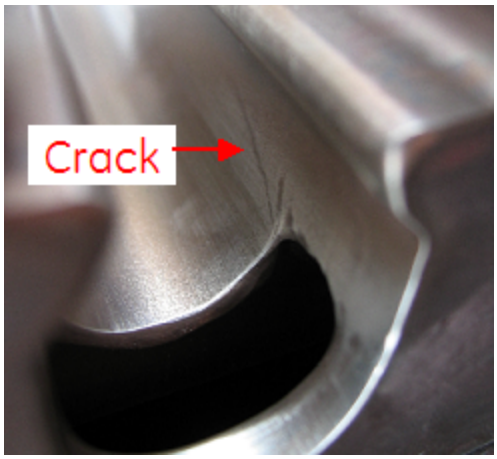


Figure 2: Crack at Broached Wheel Post and Original Cooling Slot Corner

Three cooling slot shapes exist including original, contoured, and enhanced. The contoured (released in 2002) and enhanced (released in 2006) cooling slot shapes were developed to lower stresses in the region where indications have been observed in original slot configuration wheels. Please see **Tables 1** to **6** below for eddy current inspection recommendations for original, contour, and enhanced cooling slot configuration wheels. Some rotors may have a

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

combination of enhanced, contoured, and/or original configuration wheels, depending on rotor repair history. All wheels should follow their applicable inspection recommendations. If a cooling slot crack is detected during the inspection, immediate wheel replacement will be recommended to avoid the risk of wheel post liberation during operation.

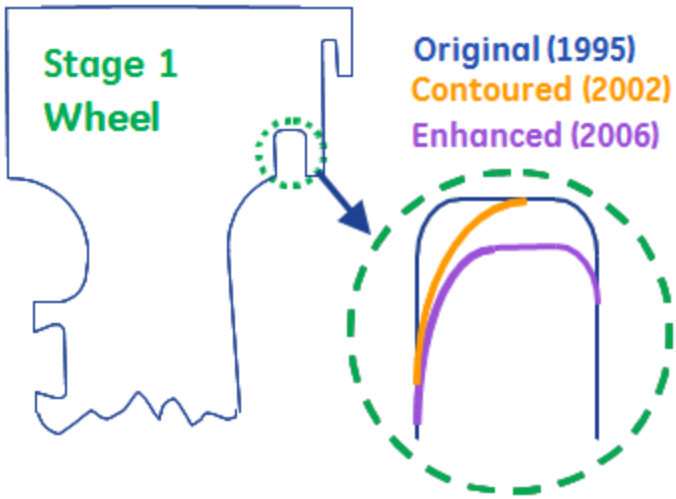


Figure 3: Original, Contoured, and Enhanced cooling slot profiles

Lock Wire Tabs

Lock-wire tab (LWT) cracks have also been detected on the turbine rotor wheels. These types of cracks can typically be repaired on site, subject to repair limits, without the need for rotor disassembly or wheel replacement.

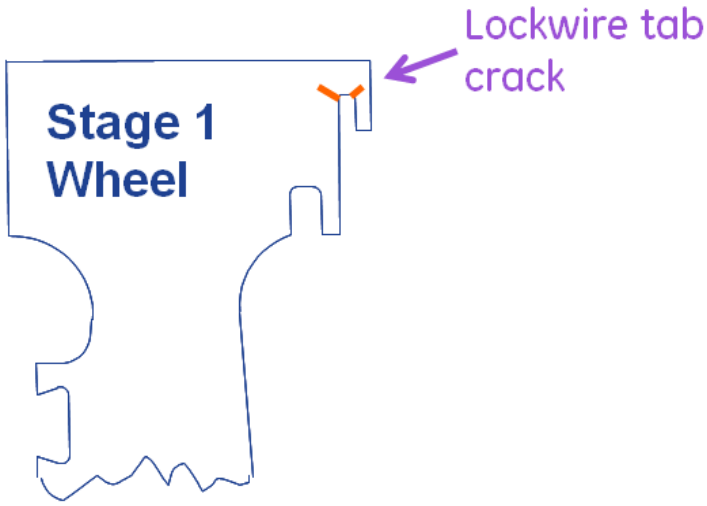


Figure 4: Potential LWT Crack Orientations

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Lock wire tab cracks may form in the upper corner of the lock wire tab groove, as shown in **Figure 4**. Cracks in this corner may be caused by either of the following:

- Damage sustained during turbine blade installation/removal
- High stresses in the lock wire fillet

GE Vernova has developed specialized procedures and tooling to help mitigate the risk of damage to the wheel lock wire tabs during turbine blade installation and removal. These operations should always be performed by GE Vernova turbine blade technicians to ensure the appropriate blade installation and removal tools and techniques are utilized. Modified or back cut turbine blades reduce turbine blade loading into the tab. Wheels manufactured today have the extra fir trees removed in order to avoid turbine blade loading in the tab altogether, as shown in **Figure 5**.

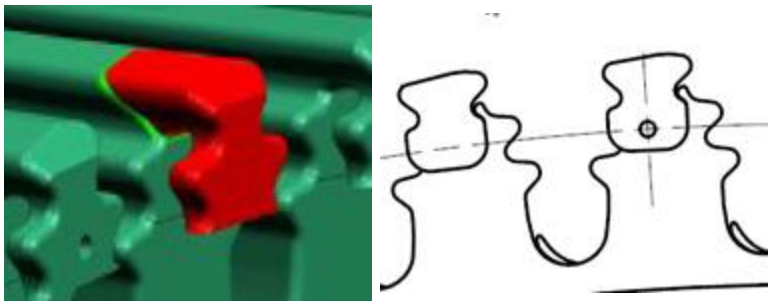


Figure 5: Lock Wire Tabs with and without Fir Tree (Modified Tab)

Balance Weight Groove

Cracking has been observed on several rotors at the balance weight groove (BWG) during routine inspections (**Figure 6**). It has been found that this cracking initiated at an adverse edge condition because of the variability in the balance weight retention method or at a location where the wheel surface was affected by an assembly or disassembly process that may have locally altered the stress state. Cracks in the forward wheel web, if left unaddressed, could result in a partial wheel material liberation, particularly when located on the stage 1 wheel.

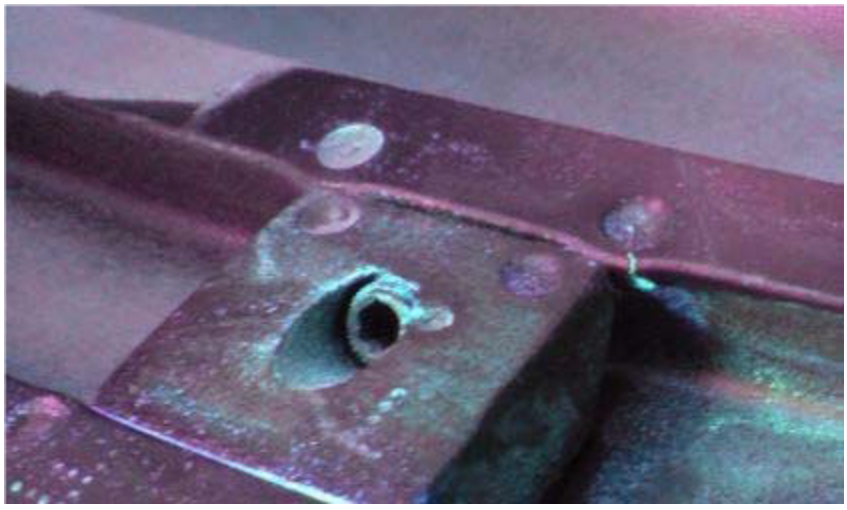


Figure 6: Balance Weight Groove Stake Mark Cracking

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Production turbine wheels have been improved, such that the balance weight groove is no longer staked or peened to deform material around the balance weight. The stakeless balance rail can be seen in **Figure 7**.



Figure 7: Stakeless BWG and Modified LWT

Material Heat Treatment

Recently, a new heat treatment - designated as three-step-age (3SA) vs two-step-age (2SA) previously - was implemented. It creates “engineered grain boundaries” that improve the hold-time fatigue capability of the turbine wheel alloy.

RECOMMENDATIONS

Event prevention and rotor health is best maintained through regular maintenance and inspections. Tables below provide recommended inspections based on the turbine rotor shot peening condition and geometric features. In addition to these inspections, it is recommended that back cut turbine blades always be installed on all turbine rotors.

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Material Heat Treatment	Peening Condition	Cooling Slot Geometry	Balance Rail & Lock Wire Tab	Recommendations		Applicable Part Numbers	
				Inspection	Interval	Frame	Stage 1
2 Step Age	Fired without Shot Peening	Original	Original	ECI Visual BI FPI UT	every 24 khrs or 900 starts	7F	227C5762 103E5795 188D7369 109E3085 101E8627 109E3896 103E5773
	Peening Modification	Original	Original	May have been peened prior to first fire. Contact your GE Vernova representative with your turbine serial number to determine the applicable peening condition. Until validated by your GE Vernova representative, assume Fired without Shot Peening recommendations apply.		7F	193D2053 193D2063
	Never Fired without Shot Peening	Original	Original	ECI	every 48 khrs or 2,400 starts [every 24 khrs or 900 starts if back cut buckets not installed]	7F	243C1494 109E5612 188D7996 114E1228 196D1915 116E2434 199D3931 117E5673 109E5303 119E2398 109E5589 119E2621
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
		Contoured	Original	ECI	at 96 khrs or 3,600 starts [every 48 khrs or 2,400 starts if back cut buckets not installed]	7F	116E3967 119E3771 119E4181
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
		Enhanced	Original	ECI	at 96 khrs or 3,600 starts	7F	323E2241
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
			Stakeless & Modified Tab	ECI	at 96 khrs or 3,600 starts	7F	144E7560
				FPI	LWT : every Major		
3 Step Age	Never Fired without Shot Peening	Enhanced	Stakeless & Modified Tab	FPI	LWT : every Major	7F	146E3626

Table 1: Applicable 7F Stage 1 Turbine Wheel Configurations

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Material Heat Treatment	Peening Condition	Cooling Slot Geometry	Balance Rail & Lock Wire Tab	Recommendations		Applicable Part Numbers	
				Inspection	Interval	Frame	Stage 2
2 Step Age	Fired without Shot Peening	Original	Original	ECI Visual, BI FPI UT	every 24 khrs or 900 starts	7F	227C5763 103E5796 101E8629 108E4197 103E5775 109E3897
	Peening Modification	Original	Original	May have been peened prior to first fire. Contact your GE Vernova representative with your turbine rotor serial number to determine the applicable peening condition. Until validated by your GE Vernova representative, assume Fired without Shot Peening recommendations apply.		7F	193D2055 193D2064
	Never Fired without Shot Peening	Original	Original	ECI	if S2B P/N 118T8421 or 122T9656 installed, inspect at 96 khrs or 3,750 starts [inspection window is 80-104 khrs] at first Major after TW2 Life Extension	7F	193D2400 199D3925 109E5591 109E5614 114E1229 116E2436
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
		Contoured	Original	ECI	at first Major after TW2 Life Extension	7F	116E3968 119E6503
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
		Enhanced	Original	ECI	at first Major after TW2 Life Extension	7F	323E2240
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
			Stakeless & Modified Tab	ECI	at first Major after TW2 Life Extension	7F	144E7562
				FPI	LWT : every Major		
3 Step Age	Never Fired without Shot Peening	Enhanced	Stakeless & Modified Tab	FPI	LWT : every Major	7F	146E3601

Table 2: Applicable 7F Stage 2 Turbine Wheel Configurations

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Material Heat Treatment	Peening Condition	Cooling Slot Geometry	Balance Rail & Lock Wire Tab	Recommendations		Applicable Part Numbers	
				Inspection	Interval	Frame	Stage 1
2 Step Age	Fired without Shot Peening	Original	Original	ECI Visual, BI FPI UT	every 24 khrs or 900 starts	9F	101E2274 103E3969 109E3982
	Peening Modification	Original	Original	May have been peened prior to first fire. Contact your GE Vernova representative with your turbine rotor serial number to determine applicable peening condition. Until validated by your GE Vernova representative, assume Fired without Shot Peening recommendations apply.		9F	227C5743 193D2050 188D7864 193D2422 193D2023 196D1632 193D2036 198D1211
	Never Fired without Shot Peening	Original	Original	ECI	every 48 khrs or 2,400 starts [every 24 khrs or 900 starts if back cut buckets not installed]	9F	109E5254
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		111E3299
		Contoured	Original	ECI	at 96 khrs or 3,600 starts [every 48 khrs or 2,400 starts if back cut buckets not installed]	9F	119E6594
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
		Enhanced	Original	ECI	at 96 khrs or 3,600 starts	9F	133E8667
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
			Stakeless & Modified Tab	ECI	at 96 khrs or 3,600 starts	9F	146E2874 113T0325
				FPI	LWT : every Major		

Table 3: Applicable 9F Stage 1 Turbine Wheel Configurations

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Material Heat Treatment	Peening Condition	Cooling Slot Geometry	Balance Rail & Lock Wire Tab	Recommendations		Applicable Part Numbers	
				Inspection	Interval	Frame	Stage 2
2 Step Age	Fired without Shot Peening	Original	Original	ECI Visual, BI FPI UT	every 24 khrs or 900 starts	9F	101E2276 103E3971 109E3893
	Peening Modification	Original	Original	May have been peened prior to first fire. Contact your GE Vernova representative with your turbine rotor serial number to determine applicable peening condition. Until validated by your GE Vernova representative, assume Fired without Shot Peening recommendations apply.		9F	188D7848 193D2106 188D7895 193D2025 193D2037
	Never Fired without Shot Peening	Original	Original	ECI	if S2B P/N 119T9404 or 122T0124 installed, inspect at 96 khrs or 3,750 starts [inspection window is 80-104 khrs] at first Major after TW2 Life Extension	9F	109E5256 111E3298
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
		Contoured	Original	ECI	at first Major after TW2 Life Extension	9F	119E6597
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
		Enhanced	Original	ECI	at first Major after TW2 Life Extension	9F	133E8776
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
			Stakeless & Modified Tab	ECI	at first Major after TW2 Life Extension	9F	146E1078 146E2875 113T0328
				FPI	LWT : every Major		

Table 4: Applicable 9F Stage 2 Turbine Wheel Configurations

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Material Heat Treatment	Peening Condition	Cooling Slot Geometry	Balance Rail & Lock Wire Tab	Recommendations		Applicable Part Numbers	
				Inspection	Interval	Frame	Stage 1
2 Step Age	Never Fired without Shot Peening	Original	Original	ECI	every 48 khrs or 2,400 starts [every 24 khrs or 900 starts if back cut buckets not installed]	6F	109E9194
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
		Contoured	Original	ECI	at 96 khrs or 3,600 starts [every 48 khrs or 2,400 starts if back cut buckets not installed]	6F	116E3992
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		323E1860
		Enhanced	Original	ECI	at 96 khrs or 3,600 starts	6F	137E1002
				BI	every HGPI when CDC is not removed		
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
			Stakeless & Modified Tab	ECI	at 96 khrs or 3,600 starts	6F	141E4569 121T7210
				FPI	LWT : every Major		
3 Step Age	Never Fired without Shot Peening	Enhanced	Stakeless & Modified Tab	FPI	LWT : every Major	6F	131T7543

Table 5: Applicable 6F Stage 1 Turbine Wheel Configurations

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Material Heat Treatment	Peening Condition	Cooling Slot Geometry	Balance Rail & Lock Wire Tab	Recommendations		Applicable Part Numbers	
				Inspection	Interval	Frame	Stage 2
2 Step Age	Never Fired without Shot Peening	Original	Original	ECI	if S2B P/N 120T1387 or 123T0809 installed, inspect at 96 khrs or 3,750 starts [inspection window is 80-104 khrs] at first Major after TW2 Life Extension	6F	109E9196
				FPI	LWT : every HGPI when buckets are removed BWG : every HGPI		
		Contoured	Original	ECI	at first Major after TW2 Life Extension	6F	116E3994
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		323E1916
		Enhanced	Original	ECI	at first Major after TW2 Life Extension	6F	137E1062
				FPI	LWT : every HGPI when buckets are removed BWG : every Major and when CDC is removed during HGPI		
			Stakeless & Modified Tab	ECI	at first Major after TW2 Life Extension	6F	141E4586 126T7211
				FPI	LWT : every Major		
3 Step Age	Never Fired without Shot Peening	Enhanced	Stakeless & Modified Tab	FPI	LWT : every Major	6F	131T7549

Table 6: Applicable 6F Stage 2 Turbine Wheel Configurations

Stage 3 Wheel and Turbine Aft Shaft Inspections

For the stage 3 turbine wheel (TW3) and turbine aft shaft (TAS), the following inspections apply regardless of frame size or part numbers:

1. TW3 balance weight grooves
    1. If staked: FPI every Major Inspection (MI) and when the Exhaust Frame is removed during Hot Gas Path Inspection (HGPI)
    2. If stakeless: no inspections necessary
2. TW3 lockwire tabs
    1. If original: FPI every HGPI while buckets are removed
    2. If modified: FPI every MI
3. TAS balance weight grooves
    1. If staked: FPI every MI and when the Exhaust Frame is removed during HGPI
    2. If stakeless: no inspections necessary

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024

Visual, Borescope (BI), and Ultrasonic (UT)

A visual inspection of all exposed turbine rotor surfaces is recommended for all non-shot peened rotors; however, it is a good practice for all users to evaluate the surface of their rotor when available. In order to complete a visual inspection, observe and borescope all of the accessible rotor surfaces. Specifically, look for overall surface conditions such as dents, dings, nicks, scratches, crack-like indications, raised material, etc.
Key features to examine:

- Broach slot fillets, slot bottoms, tangs, and cooling slots (Visual)
- Turbine blade cooling slots in wheel broach slots (Visual/BI)
- Spacer rim seal tooth rubs/coating (Visual)
- Balance weights: looseness/migration, staking quality, tight set screw (Visual/BI)
- Nut damage or surface indications in flanges and webs (BI)
- Counterbore outside diameter ligaments on distance piece and turbine aft shaft (UT)

Fluorescent Penetrant Inspection

Fluorescent penetrant inspection (FPI) is recommended to detect cracks on balance weights and lockwire tabs. However, it is not recommended for the wheel posts, as cracks under the dovetail tangs are typically closed.
Recommended features to inspect:

- Staked balance weight grooves: at all major inspection intervals and when the CDC is removed during a HGPI.
- Lock wire tab fillets: When turbine blades are removed during routine maintenance intervals.

If crack-like indications are found, please contact your local GE Vernova service representative for disposition. In most cases, an in-situ repair procedure can be used to remove cracked lock wire tabs or balance weight stake marks. A few isolated instances have occurred where wheel replacement was necessary as the repair limits were exceeded.

Eddy Current Inspection

GE Vernova has developed an ECI method with advanced detection devices and techniques for turbine dovetail inspection (**Figure 8**). This method is currently the most effective non-destructive test (NDT) for identifying indications in the wheel dovetails.

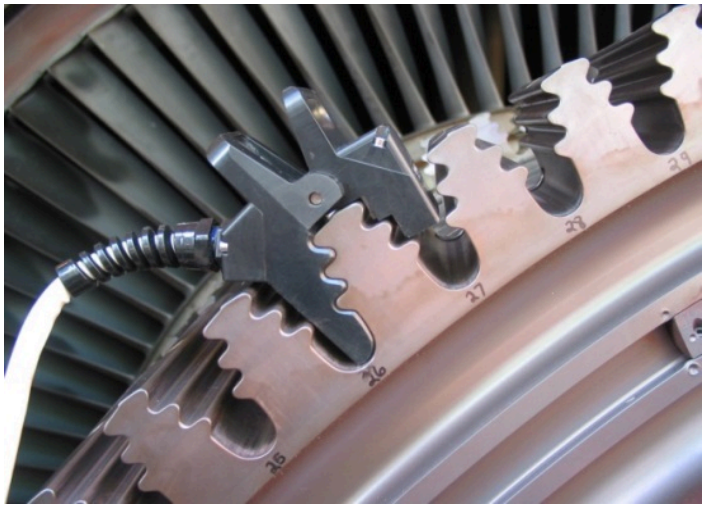


Figure 8: Eddy current inspection

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Dovetail ECI is recommended for all wheel posts until rotor retirement at the frequency defined in **Tables 1-6**. Third stage wheel posts and turbine wheels with any cooling slot configuration may also be inspected with the same equipment, if elected.

Prior to the ECI, the turbine rotor must be appropriately cleaned to facilitate proper functioning of the ECI equipment and accurate readings. A preferred method for cleaning is by dry ice (CO_2) blasting. This cleaning typically requires a single 12-hour shift to clean the rotor after turbine blade removal, and also requires the surfaces be free of oil beforehand. In the event that CO_2 blast is not available, a more arduous and lengthy hand-cleaning process is acceptable (2-3 shifts). The rotor inspection then takes approximately three shifts (~32 hours) to complete Stages 1 and 2. Both the rotor cleaning (performed by qualified vendor) and the turbine rotor inspection can be scheduled through your local GE Vernova representative.

Modified Stage 1 and 2 Turbine Blades

The potential for crack initiation at the cooling slot location and lock wire tabs can be improved by reducing the amount of turbine blade load passing through this area of the wheel. GE Vernova has developed a turbine blade modification to change the load distribution in the wheel post, thereby decreasing the stress level at the cooling slot. This modification is accomplished by machining a tapered relief into the turbine blade dovetail pressure faces adjacent to the cooling slot (see **Figure 9**). This relief is often referred to as a dovetail “back cut”. Relief cut modifications have been incorporated into new production F-class turbine blades.

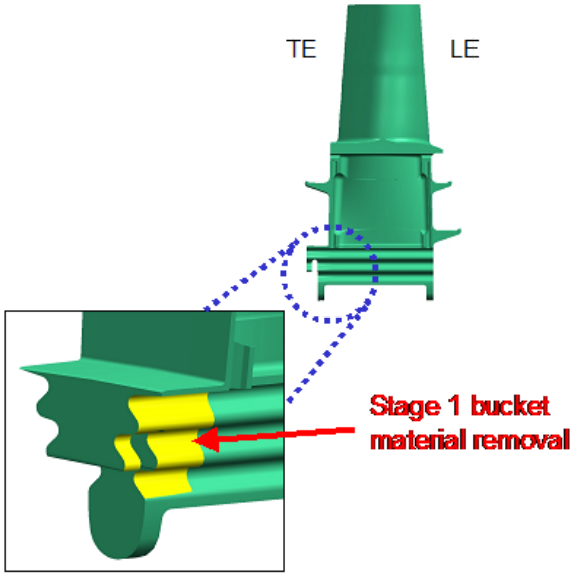


Figure 9: Turbine blade dovetail modification

The “back-cut” modifications are also performed as a part of the standard turbine blade repair/refurbishment process and should be applied to any inventory parts by a GE Vernova Service Center. It is recommended to apply dovetail modified Stage 1 and 2 turbine blades at the next scheduled maintenance (hot gas path inspection or major inspection).

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

SCOPE OF WORK

Turbine blades should be removed prior to beginning these inspections per hot gas path outage scope. CO2 blast cleaning of turbine rotor can be completed in one shift (12 hours), including a prior hand wiping to remove oil from the dovetail surfaces if necessary. ECI of first and second stages can be completed in three shifts, or all three stages in four shifts. Total time between turbine blade removal and installation should be four shifts (two days). These durations are estimates only.

TIL COMPLETION CRITERIA

This TIL is considered complete when the turbine wheels reach end of life and are removed from service.

TIL DISPOSITION

Disposition of TILs should be entered in local records and also in GE Vernova ServiceNow. Follow the below instructions for entering the disposition record;

- Log into the Power ServiceNow at https://gepowerpac.service-now.com/til_new/ using your GE Vernova SSO number and password.
- Select "TIL Disposition".
- Click on the TIL for the serial number you want to update.
- Choose the most appropriate "Disposition Status" and enter "Disposition Notes".
- Click "Save".

Contact your local GE Vernova Services representative for assistance or for additional information

© 2024 GE Vernova

The proprietary information published in this Technical Information Letter is offered to you by GE Vernova in consideration of its ongoing sales and service relationship with your organization. However, since the operation of your plant involves many factors not within our knowledge, and since operation of the plant is in your control and ultimate responsibility for its continuing successful operation rests with you, GE Vernova specifically disclaims any responsibility for liability based on claims for damage of any type, i.e. direct, consequential or special that may be alleged to have been incurred as result of applying this information regardless of whether it is claimed that GE Vernova is strictly liable, in breach of contract, in breach of warranty, negligent, or is in other respects responsible for any alleged injury or damage sustained by your organization as a result of applying this information. This Technical Information Letter contains proprietary information of GE Vernova and is furnished to its customer solely to assist that customer in the installation, testing, operation and/or maintenance of the equipment described. This document shall not be reproduced or distributed in whole or in part nor shall its contents be disclosed to any third party without the written approval of GE Vernova. All rights reserved.

Date Printed: 15 JULY 2024